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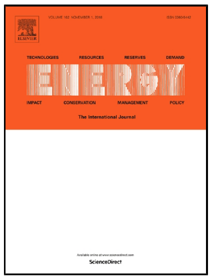

Long-Term Electricity Demand Forecast and Supply Side Scenarios for Pakistan (2015-2050): A LEAP Model Application for Policy Analysis

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Long-Term Electricity Demand Forecast and Supply Side Scenarios for Pakistan (2015-2050): A LEAP Model Application for Policy Analysis

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Abstract

Pakistan is facing electricity crises owing to lack of integrated energy planning, reliance on imported fuels for power generation, and poor governance. This situation has challenged governments for over a decade to address these crises. However, despite various conformist planning and policy initiatives, the balance between demand and supply of electricity is yet to be achieved. In this study, Long-range Energy Alternatives Planning System (LEAP) is used to develop Pakistan's LEAP modeling framework for the period 2015-2050. Following demand forecast, four supply side scenarios; Reference (REF), Renewable Energy Technologies (RET), Clean Coal Maximum (CCM) and Energy Efficiency and Conservation (EEC) are enacted considering resource potential, techno-economic parameters, and CO_2 emissions. The model results estimate the demand forecast of 1706.3 TWh in 2050, at an annual average growth rate of 8.35%, which is 19 times higher than the base year demand. On the supply side, RET scenario, although capital-intensive earlier in the modeling period, is found to be the sustainable electricity generation path followed by EEC scenario with the lower demand of 1373.2 TWh and minimum Net Present Value (NPV) at an aggregate discount rate of 6%. Conclusion section of the paper provides the recommendations devised from this study results.

Keywords: Electricity crises, Electricity demand and supply, Energy and Power Policy, LEAP, National climate change policy, Pakistan

1. Introduction

The electricity crises of Pakistan are reflective of failed energy planning and policy regime of the country alongside poor governance in the sector. The situation is so much grave that at the moment shortage of electricity is not the only problem to be tackled but other issues like expensive electricity, the circular debt of the energy sector, and higher Transmission and Distribution (T&D) losses hold the true face of Pakistan's power sector. Moreover, despite being ranked seventh among most vulnerable countries to the climate change, Government of Pakistan (GoP) is still ambiguous to undertake implementation of appropriately developed environment and climate change policy [1].

The electricity crises of the country which surfaced during the year 2006 can be traced way back in the 1990s when ambitious power policies were announced by the various governments and thereby seeking investment from the private sector [2, 3]. Most of the power policies developed then were without any formal planning studies undertaken using energy modeling tools [2]. It is despite the fact that several modeling tools are successfully utilized globally for devising energy and power policies. Various developing countries such as; Bangladesh China, Crete, India, Greece, Ghana, Turkey, Nigeria, and others have addressed critical energy issues by assisting effective policy formulation using these tools [4]. In the contemporary literature too, several studies on energy modeling tools are undertaken for various countries as summarized in **Table 1** [5-25]**.**

However, there is a dearth of such studies undertaken in Pakistan. A summary of limited relevant studies of Pakistan, which utilized energy modeling tools, is given in **Table 2** [26-29]. It also provides the scope and considerations covered or otherwise in these past studies and appropriately addressed in this study. This study, therefore, includes the maximum considerations pertaining the electricity sector of Pakistan and further emphasizes a future direction for a sustainable energy system of Pakistan by recommending a comprehensive IEP effort to devise rational energy policy and keeping such endeavours live to incorporate future adjustments.

In terms of energy economic dynamics, Pakistan's power sector is also facing the worst ever financial crises of its history with a soaring circular debt of USD 7.6 billion [30]. The demand-supply gap of electricity had alone during the year 2010 caused a loss of 2.5% in the overall GDP leading to unemployment of more than half million workers in the industrial sector [26]. One of the reasons for the increase in the electricity demand and thus the demand-supply gap is increased population, rapid urbanization, improved living standards and rural electrification [27]. However, the plans and policies of the past could not deliver to meet the ever-increasing electricity demand. As such, it is realized that meeting electricity demand is mutually related to Integrated Energy Planning (IEP), and this planning is almost ineffective without using energy modeling tools and appropriate implementation of planning and policy narrative derived from IEP effort.

Globally, following the oil crises of the 1970's, policymakers had not only emphasized on identifying reliable and efficient energy sources but focus also shifted towards ensuring sustainable energy pathways for the future [31]. As such, energy scenario planning has become a most popular approach to project future demands and alternate supply pathways to meet the demand. However, it is essential to understand that effective energy planning does not conclude at developing a model of any specific energy system. Instead, it should assess and project various future energy patterns and suggest policy options by keeping model alive for a rational policy formulation to achieve sustainability. So far, Pakistan lacks in the appropriate application of such planning paradigm and policy formulation. Meanwhile, segregated energy planning and diversified policies at various levels have pushed the country into severe energy crises.

This study focusing the electricity supply sector as one of the important modules of IEP undertakes the electricity demand forecast and simulate four supply side scenarios using the LEAP model over the study period 2015-2050. The electricity demand projections are linked with the demographic and socio-economic parameters. The supply side scenarios depict the future pathways to provide policy makers with an insight into the system impacts with respect to pursuing these pathways or otherwise; elaborate details on resources utilization, conversion processes, cost parameters, and CO_2 emissions under each scenario. The four supply-side scenario alternatives of the study are; Reference (REF), Renewable Energy Technologies (RET), Clean Coal Maximum (CCM) and Energy Efficiency and Conservation (EEC). Analyzing the impacts of each of these scenarios from a policy perspective help to examines the interactions that occur when several technologies under a single policy are considered in different time horizons. As such, this study helps in answering the following key policy questions.

- What is the future electricity demand of the country and how it can be satisfied in the medium to long-term?
- What could be future electricity supply pathways for the county?
- How would each of future electricity supply pathway impact cost and environment?

LEAP model used in this study is one of the widely used energy modeling tool by various organizations in more than 190 countries including government departments. LEAP modeling effort requires low initial data and have built-in "Technology and Environmental Database" for greenhouse gas (GHG) mitigation assessment [2].

The results of this study provide the estimated electricity demand forecast and insight into future options for electricity generation for the period 2015-2050. Based on the results of this study, it is recommended to undertake IEP development by the government for policy analysis towards devising appropriate programs and projects to achieve sustainable supplies for meeting the future demand.

This paper in section 2 provides analyses of the power sector of Pakistan. The key power and climate change policies of the country are briefly discussed in section 3. In section 4 LEAP model methodology for demand forecast and scenario development, as well as the main assumptions and data input to the model, has been discussed in detail. The model results pertaining demand forecast, sensitivity analysis and scenario development are presented in section 5. The detailed discussion of the results is given in section 6, and finally, the conclusion and policy recommendations based on this study are summarised in section 7 of the paper.

2. Electricity Demand-Supply situation and Exiting Policies of Pakistan

The electricity demand-supply situation of Pakistan alongside relevant energy policies are briefly discussed as follows.

2.1 Electricity Demand Situation

Pakistan has a population of 207.774 million as per recent census of 2017. This census also reveals that 36.34% of the total population lives in urban areas, while 63.66% is living in remotely located rural areas [32]. The total number of households in the country are 32.2 million, and an average household has 6.4 members [32]. The rise in population has also caused the increase in electricity demand. However, the electricity supply side growth has had been at a snail's pace. There are five major electricity consumer groups in the country, which are: domestic; industrial, commercial, agricultural and others such as bulk supplies, street lighting, and other electricity consuming public services offered by the government. The domestic sector leads all other sectors with 40.8 TWh (45%) electricity consumption followed by industrial, commercial, agricultural, and others which consumed 24.9 TWh (28%), 6.4 TWh (7%) , 8 TWh (9%) and 10.2 TWh (11%) of electricity respectively in the year 2015 [33]. It is significant to note that with the beginning of the 21st century, domestic sector alone has posted average annual growth of 5.94% in the electricity consumption, whereas total consumption by all the consumer groups has increased at an annual average growth rate of 4.54% [33].

Further, the number of electricity consumers in domestic sectors has also grown at the rate of 2% per annum followed by industrial, commercial, agricultural and others by 2.7%, 3.17%, 4.29%, and 2% respectively [34]. **Table 3** [35] shows sector-wise growth in the number of electricity consumers over the past five years.

The rapid growth in the number of electricity consumers and thus in the electricity demand has been hardly met until 2006. As such, in the following years, there has had been electricity shortfall causing load shedding across the country. The patterns of electricity supply and demand from 2006 to 2016 with a historical account of load shedding in the country are shown in **Fig. 1** [36].

NTDC manages the schedules and patterns of load shedding when a gap exists between demand and supply. Subsequently, a load management plan is devised in accordance with the generation capacity.

It is, therefore, apparent that although electricity consumption and thus demand in the country has grown over the past many years, yet the electricity consumption per capita is only 452 kWh which is around one-fourth of world average [28]. Further, around 51 million of the total population of the country has no access to electricity mainly due to limited generation capability and minimum expansion in T&D network alongside higher T&D losses [37]. As such, Pakistan needs to develop an integrated energy planning and implementation framework to address the electricity shortfall and enhance access to the commodity.

2.2 Electricity Supply Situation

Pakistan has limited indigenous reserves of conventional energy resources (oil and gas), as such, need to rely on import of these resources to meet its energy demand. However, there are abundant resources of coal reserves estimated to be 175 billion tons in the Thar desert of Sindh province which seek greater attention for harnessing through efficient technologies [38]. The renewable energy resources of the country are even in greater abundance and could form a substantial share in the overall energy mix for the power generation. However, Pakistan's energy mix for electricity generation of 106.97 TWh in 2015 is broadly dominated by oil and followed by hydropower, natural gas, nuclear, wind, bagasse, and coal respectively as shown in **Fig. 2** [39].

It is unfortunate that county's renewable energy resources so far could not be harnessed to any optimal level even though prices of renewable energy technologies have considerably been reducing over the years. **Table 4** [39] shows the total installed capacity of various entities of Pakistan's power sector which is mainly dominated by thermal power generation by the private sector Independent Power Producers (IPPs).

A detailed electricity flowchart of Pakistan is shown in **Fig. 3** which illustrates the overview of primary energy resources consumption for electricity generation in ktoe (Thousand tons of oil equivalent), total TWh generated, T&D losses, and TWh electricity consumption by different consumer groups for the year 2015.

The electricity supply situation of Pakistan is not only indicative of poor energy mix for the power generation but at the same time shows lack of expansion in the supply side capacities as well as efforts to reduce the T&D losses to an acceptable level. The cause of such a situation is the absence of the effective planning, applicable policy, and well-defined framework to ensure sustainable electricity supplies.

2.3 Pakistan's Power and Climate Change Policies

The integrated energy planning and policy is indispensable for the overall development and economic growth of developing countries like Pakistan. Unfortunately, Pakistan does not have an integrated energy and power policy even after seventy years of its independence. The never-ending restructuring and complexity of government organizations with vague roles and responsibilities keeps Pakistan away from developing an integrated energy and power policy towards achieving sustainable development. At the same time, there are varied and disintegrated policies announced by GoP such as; petroleum policy, power policy, renewable energy policy, energy conservation and efficiency policy and climate change policy. There is hardly any integration and uniformity in these policy documents announced from time to time. The power policies of Pakistan are developed and implemented mainly by Power Division of newly created MOE. GoP in 1994 announced first-ever power policy, followed by various policy endeavours announced in the next 20 years. However, none of the policies announced so far is based on formal energy planning studies undertaken from time to time except that demand forecasts are generally used to estimate future fuel supplies alone. Thus, due to erroneous preferences, lack of modeling exercises and implementation issues, Pakistan could not witness substantial development across various sectors of the economy as per set targets [2]. A summary of various energy and power policies announced so far by the GoP with key features, focus, and targets of each policy are given in **Table 5** [2]**.**

In addition to various shortcomings of energy and power policies announced by GoP, the environmental concerns of power generation were rarely given any importance in these policies. Although, in February 2013 GoP had also announced the first-ever National Climate Change Policy (NCCP) [40]. This policy focuses on ensuring water, food, and energy security by integrating climate change issues into other inter-related national policies. Further, the emphasis has been laid on strengthening inter-ministerial decision-making and coordination mechanism on climate change. Nonetheless, the fate of NCCP like other policies is only contained to be a document with no implementation in sight.

In the meantime, Pakistan has also submitted ambitious Intended Nationally Determined Contributions (INDCs) as per the Paris Agreement of 2015. As per submitted INDCs, the estimated GHG emissions of Pakistan in 2030 would be 1,603 Mt.CO_2 equivalent, which is further expected to increase for many decades before any decrease in the emissions. A reduction of up to 20% in the projected emission figures for the target year 2030 would require a total investment of approximately USD 40 billion at the current price. Further, Pakistan's adaptation needs ranges between USD 7 to USD 14 billion/annum during this period [41].

In the light of above, energy and power policies of Pakistan based on IEP should not only ensure energy security but at the same time address the environmental concerns from electricity generation.

3. Methodology and Data

This exploratory study focuses on the electricity supply system in Pakistan and using LEAP model forecasts electricity demand for the period 2015-2050 as well as develop four supply side scenarios. LEAP is a user-friendly energy modeling tool which is globally used for the analysis of energy policy and assessment of climate change mitigation. LEAP facilitates a scenario-based modeling approach to track energy resource extraction, production, and consumption in all sectors of the economy. Lower initial data requirement of the LEAP model and built-in technology and environment database suits development of Pakistan's LEAP modeling framework.

3.1 LEAP Modeling Framework for Pakistan

LEAP with its energy accounting capabilities matches demand with supply side electricity generation and also provides the system impacts including electricity generation by sources, the requirement of installed capacity, cost parameters and environmental emissions. As such, following Pakistan's electricity demand forecast, four supply side scenarios have been developed in this study. While developing Pakistan LEAP modeling framework, 2015 has been set as the base year and 2050 as the end year. The first supply side scenario is Reference (REF) scenario which is an energy pathway as per GoP's current plans and policies. The Renewable Energy Technologies (RET) pathways envisage maximum penetration of renewable energy and Clean Coal Maximum (CCM) scenario foresee the use of efficient technologies for power generation. The fourth scenario is named as Energy Efficiency and Conservation (EEC) scenario which focuses on energy conservation objectives, thereby reducing electricity consumption and thus reduced demand. In each of these scenarios, in addition to the projection of generation and installed capacity requirements CO_2 emission and NPV under different discount rates have been estimated as well.

In the LEAP, the input data set consists of various modules such as key assumptions, demand, transformation, and resources. The key assumption module includes Gross Domestic Product (GDP), GDP growth, total population, population growth, number of consumers and their growth alongside other relevant parameters. The demand module is divided into five sectors; domestic, industrial, commercial, agricultural, and others which are linked to demographic and socio-economic parameters to forecast electricity demand. **Fig. 4** illustrate the electricity demand forecast structure of the LEAP model.

In the transformation module, energy transformed from energy sources as input to energy product is modeled using a range of electricity generation technologies which includes those in operation during the base year and other anticipated in government's plans as well as considered in accordance with the emerging trends of the generation technologies in the future. **Fig. 5** shows the electricity generation module for Pakistan in the LEAP model.

For each technology type, dispatch rule, process efficiency, historical production, exogenous capacity, lifetime, the retirement of the plant, maximum availability, capital cost, O&M, and variable costs are considered. The total cost associated with the electricity transformation module in the LEAP model is a function of total cost, unit price of energy source input, fixed and variable costs of electricity generation of relevant technologies expressed in terms of NPV.

3.2 Main Assumptions and Data Input in Model

The electricity demand projections are globally linked with the demographic and socio-economic parameters. As such, for this modeling exercise, the most reliable studies and reports were reviewed to develop a consistent set of data such as; overall GDP, sector-wise GDP, GDP growth rates, sector-wise electricity consumption, number of consumers, population, and households. **Table 6** [35, 42] summaries the key assumptions and model input parameters and their growth rates for the demand module.

The overall electricity demand is based on the demand of the five consumer groups; domestic, industrial, commercial, agricultural, and others. In this context, electricity demand data of the last 25 years (1991-2015) have also been analyzed and used as input in the demand module. The summary of electricity demand data of the last 25 years for each consumer group is shown in **Fig. 6.**

The electricity consumers growth rate and electricity intensity data have been taken from NTDC and NEPRA reports [36]. The process efficiency of renewable technologies (solar and wind) has been assumed to be 100 percent. However, the electricity generated from the renewable energy technologies is considered less than its actual potential due to seasonal variations. The capital cost, operation, and maintenance cost (fixed and variable) of different technologies are taken from Energy Information Administration [43], the USA report of 2013 [43]. However, no report of EIA or International Energy Agency (IEA) pertaining the cost data of power plants is available for the based year of this study. Thus, the capital costs for all existing plants, operation and maintenance costs (fixed and variable) are calculated from EIA 2013 report for the year 2015 as per the method of inflation [43]. Exogenous capacity and historical production of each power plant are also added in the LEAP model which were also taken from NTDC and NEPRA reports [36]. The exogenous inputs and endogenous variables which are used in the Pakistan LEAP model are given in **Table 7.**

The industrial and agricultural GDP growth rates of 5.81% and 2.73% and that of overall GDP growth of 5% has been considered during the study period [44]. Since the GDP data for the other consumer groups is not available, as such, for modeling, these sectors are linked to the overall GDP growth rate.

Table 8 [37, 45-47] summarizes the detailed data, for the base year 2015, pertaining power generation, installed generation capacity, the amount of fuel consumed, electricity generated per source alongside fixed and variable costs as well as efficiency and availability in percentages utilized in the LEAP model of this study.

The LEAP model low data requirements and modest structure, on the one hand, are advantageous to commence the modeling exercise but on the other hand, have limitation pertaining decision making which is mainly relying on modeler preferences. As such, training and expertise in energy modeling is a pre-requisite to work on such bottom-up energy models. Nevertheless, this and similar modeling efforts on any scale for a developing country like Pakistan should be viewed as significant groundwork to move forward for more IEP level studies using various modeling tools.

4. Scenario Development

In the electricity planning, the supply side scenarios depict the future pathways which provide policy makers with an insight into the system impacts with respect to pursuing these pathways or otherwise; elaborate details on resources utilization, provide cost parameters, and GHG emissions under each scenario. This approach allows assessing the impacts of individual policy and examines the interactions that occur when multiple pathways under a policy are under consideration in different time horizons.

In summary, the key purpose of scenario development is to analyze:

- Possible future events by considering possible alternative outcomes.
- Assessment of energy systems under various assumptions about the economy
- Planning through modeling the interaction between population, economy, emissions and energy resources

In this study, four supply side scenarios have been developed to meet the demand forecast for the study period (2015-2050). A summary description of the four scenarios of the study is given in **Table 9.**

More elaborate description of each of these four scenarios of this study follows as under:

4.1 Reference (REF) Scenario

REF scenario portrays the supply side pathway for the period 2015-2050 as per policy and plans followed by the government under Pakistan Vision 2025 and National Power System Expansion Plan (NPSEP) of 2011. Energy projects planned under China Pakistan Economic Corridor (CPEC) project are also taken care of under this scenario. It is expected that no major change in the on-going policy regime will be made and development trends will continue. The most important perspective of existing government plans is that of not adding any new oil-based plants in future power generation is also duly taken care of in this scenario. Moreover, the underlying fact that gas shortage is expected to get more severe in the study period which may elevate the energy crises is also considered [48]. All these inputs are available maximum up to 2030, as such, the same trends are followed in the study for the remaining modeling period.

4.2 Renewable Energy Technologies (RET) Scenario

The vision of this energy pathway is to inject more share of renewable energy and introduce relevant technologies of hydropower, solar, wind, and biomass-based system in Pakistan's overall energy mix for power generation. This strategy shall reduce the share of fossil fuels so that energy and economic security could be attained. The estimated realizable potential of power generation through solar PV and thermal technologies alone is six times the total installed capacity in the country [49]. A such, taking an optimistic approach; the share of renewable energy resources has been increased gradually considering future technological developments and lowered prices so that these resources contribute around 50% in the overall energy mix for the power generation under this scenario.

4.3 Clean Coal Maximum (CCM) Scenario

In Pakistan's electricity generation mix for 2015, the share of coal is only 0.1% compared to the world's average of 41% [34, 39]. This is despite the fact that country is blessed with world's 6th largest coal reserves of 186 billion out of which 175.506 billion tons of lignite is alone estimated in the Thar district of Sindh province [39]. CCM addresses meeting electricity demand by a major share of indigenous coal by using clean coal technologies in this scenario. In this regard, technologies such as Coal based Carbon Capture and Storage (CCS) and Integrated Gasification Combined Cycle (IGCC) are also considered to lower the emissions. These technologies (IGCC and CCS) have the potential to capture/reduce 85-90% and 50-70% of the emissions respectively. An effort has also been made to take care of Pakistan's INDC submission in view of the Paris Agreement for reduction of CO_2 emissions to a level of 20% less than the base year in 2030 and 30% and 40% less than the base year in 2040 and 2050 respectively.

4.4 Energy Efficiency and Conservation (EEC) Scenario

Efficient use of energy and its conservation is very important since one unit of electricity saved at the user end reduces the need for new capacity addition by two units. As such, efficient use of energy can be achieved at less than one fifth the cost of the fresh capacity addition. The use of old and traditional technologies and less efficient appliances elevates the consumption of electricity. EEC scenario is based on the assumption that use of modern technologies and efficient appliances as well energy conservation measures shall cause the reduced electricity demand. This assumption is supported by Asian Development Bank [50] study that energy-saving potential in Pakistan is 25% in the domestic sector, 14.55% in the industrial sector, 23.86% in the commercial sector, 41.1% in the agriculture sector and 5.69% in others [50]. As such, efficiency and conservation measures of this scenario are estimated to reduce electricity demand by 20% of the total forecasted demand under reference scenario.

5. Results

Following sub-sections of the paper put forward the results of this study viz-a-viz demand and supply-side projections with associated techno-economic finding as well as CO_2 emissions.

5.1 Demand projections

The electricity demand projections from 2015 to 2050 under the reference scenario for each consumer group and overall are shown in **Table 10.** The estimated growth in electricity demand can be attributed to the rise in population and electricity consumers, rapid urbanization, improved living standards, rural electrification, and stable GDP.

The overall projected electricity demand of Pakistan in 2050 is forecasted to be 1706.3 TWh which was only 90.4 TWh in 2015. As such, a growth 19 times higher than the base year demand with an estimated annual average growth of 8.35% is projected. The existing official electricity demand forecast by NTDC is until 2035 wherein a demand of 687.517 TWh from all the consumer groups is projected [34]. Some other researchers, Gul and Qureshi [29] and Perwez, Sohail [26] have also forecasted electricity demand of 368 TWh and 312 TWh respectively during the year 2030. As such, these researcher's forecasted demand for 2030 is almost half of what has been projected by NTDC during 2035. Similarly, Ishaque [28] has an estimated electricity demand forecast for the year 2035 to be 303.7 TWh in 2035 which is also almost half the NTDC forecast for the same period [28]. It is, therefore, noted that electricity demand forecasts by NTDC are generally exorbitant thereby considering higher growth rates of the economy which have hardly been close to the factual figures.

The demand forecast for this study for the years 2030 and 2035, is 330.1 TWh and 504.4 TWh respectively which is not only commensurate with other researchers forecasted estimates but well below the NTDC forecast.

It is evident from the base year (2015) demand that each consumer group, domestic, industrial, commercial, agricultural and others have share of 45.1%, 27.6%, 7.1%, 8.9%, and 11.3% respectively in the overall demand which is forecasted to be 39.1%, 38.8%, 8.7%, 3.5%, and 9.8% respectively in the end year 2050. This study forecast, as such, estimate that demand from the domestic and industrial consumer shall be competitively higher than other consumer groups with substantial growth in the economy. This fact can also be verified by the projected number of electricity consumers which shall significantly increase during the forecast period as shown in **Table 11.**

Since REF, RET and CCM scenarios are driven by similar demand-side assumptions. Thus demand forecast for these three scenarios is same; 1706.3 TWh for the year 2050. However, in the case of EEC scenario, the projected electricity demand is estimated to be 1373.2 TWh, which is 20% less than the demand forecasted under the reference scenario.

5.1.1 Sensitivity analysis of demand forecast

In this study Sensitivity Analysis (SA) is undertaken to determine the robustness of the LEAP model demand forecast results. The SA helps in understanding that on varying certain input parameters, how it impacts the overall on the result? The SA of the electricity demand forecast of this study, as such, evaluates the impacts of imposed variations on the inputs parameters pertaining population, GDP and number of consumers on the results as follows under.

Varying the Population Growth Rate

Fig. 7 shows the increase in electricity demand with varying population growth rates at 1.9%, 2.1%, and 2.23% respectively. The population growth rate of 1.9 % has been taken from the Pakistan

Economic Survey 2015, while a further increased population growth rate assumption of 2.12% and 2.23% as expected shows further increase in demand forecast for the study period.

Varying the GDP Growth Rate

Fig. 8 shows the forecasted electricity demand at different GDP growth rates of 4.32%, 5%, and 5.5% respectively. The GDP growth rate of 4.32% and 5% have taken from NTDC reports while a 5.5% GDP growth rate is an optimistic assumption. It is evident from **Fig. 8** that variation in GDP growth rate would impact the electricity demand. At a lower GDP, the demand would be sufficiently low compared to the demand forecast at optimistically assumed higher GDP growth rate.

Varying the Number of Consumer Growth Rate

Fig. 9 shows the forecasted demand at a different growth rate for varying total number of electricity consumers. The growth rates of 14.16% and 14.65% are taken from the NTDC report, while 15% is an assumption considered under SA. The results under varying growth rate of number consumer are also significantly important.

It is concluded from the SA of demand forecast that population, GDP, and the number of consumers are independent parameters which directly affect the electricity demand and thus must be taken into consideration while taking important policy decisions.

5.2 Supply-Side Scenarios

The supply-side projections of each of four scenarios (REF, RET, CCM, and EEC) of this study comprises of the generation to meet the demand, the installed capacity to ensure the required generation, the estimated CO_2 emissions and finally the cost analysis in terms of NPV.

5.2.1 Generation

The electricity generation under all the scenarios for the study period is projected by the model using various fuel and technology mix in order to meet the demand. **Fig. 10** shows the electricity generation under all the scenarios of this study. In REF scenario, the share of oil in electricity generation mix is 35.7% in 2015 which will end up to be 0% in 2035 and so on as per existing policies of the government in order to reduce the import bill of the country. The share of large hydropower is 20.6% in 2015 which is anticipated to decrease at a level of 13.3% in 2050. Similarly, the share of run of the river which is 9.9% in 2015 shall be only 4.4% in 2050. As such, the overall reduced share of the hydropower plants by the year 2050 shall be substantiated by various other fuels and technologies. The share of solar, wind and biomass in the base year is around 0%, 1%, and 0.3% respectively which is projected to be 10.1%, 8.1%, and 3.9 % respectively in 2050. Similarly, the share of natural gas open cycle power plant is 19.6% in the base year which shall be 0% in the year 2035 and so on as per existing policies of the government so as to divert natural gas to other sectors of the economy such as industries and fertilizer units. The share of natural gas combined cycle (NGCC) power plants and coal based units were 9.2%, 0.1% respectively in 2015 which are projected to be 29.8%, 26.3% respectively in 2050. The NGCC, therefore, would form a major share of 29.8% for the electricity generation under REF scenario during 2050 followed by the coal based plants which shall generate 26.3% of total electricity in 2050.

In the RET scenario, the share of solar PV, wind and biomass resources has been projected to be 14.52%, 15.12%, 4.09% respectively in 2050. In addition, the generation share from hydropower and RoR plants is projected to be 14.11% and 5.53% respectively. The overall share of renewable energy, inclusive of hydropower, which was 33.6% in the base year 2015 is projected to be 53.4% in 2050 in this scenario.

In the CCM scenario, with no generation of electricity from oil and open cycle natural gas plants from 2035 onward, the Integrated Gasification Combined Cycle (IGCC) and Carbon Capture and Storage (CCS) technologies have been considered for the coal based power generation. The Coal based IGCC and CCS technologies have 45% and 34% efficiencies respectively and are certainly better of than old technologies used in coal power plants with lower efficiencies. Accordingly, with these technologies, the share of coal is about 23.46% in the year 2050 which shall be major contributing fuel for electricity generation in this scenario. The second major fuel for electricity generation in this scenario is hydropower with a share of 12.53% by 2050. Similarly, NGCC share comprises 11.96% of the overall electricity generation by 2050. The share of solar, wind and biomass for electricity generation under this scenario is projected to be 11.6%, 9.84%, and 2.6% respectively in 2050. Finally, the share of nuclear, RoR, Coal based IGCC and Coal based CCS have been projected to be 4%, 5.27%, 11.17%, and 7.45% respectively in overall fuel mix for electricity generation in 2050.

In the EEC scenario, the electricity demand (1373.2 TWh) is 20% less as compared to other scenarios in the end year of the study period. The lower electricity demand requires less electricity generation which will help to address the electricity deficit and ultimately saving huge capital and other costs alongside reduced emissions. In this scenario, the share of coal for the electricity generation is around 28.45% until the end year which makes it leading fuel in the electricity generation by 2050. The second major share in electricity generation is projected to be from NGCC which shall form 26.14% by 2050. Similarly, hydropower share in electricity generation is projected to be 13.58% in the year 2050 which shall make it the third largest contributing fuel for the electricity generation. The share of solar, wind and biomass is projected to be 11.7%, 9.6%, and 3.9% respectively in 2050. The shares of nuclear and RoR are estimated to be 3.1% and 3.45% respectively in 2050 under this scenario.

5.2.2 Installed capacity

The installed capacities of various types of power plants to ensure the required generation in each of the four scenarios are shown in **Fig. 11.**

The major installed capacities of power generation would come from NGCC and coal-based electricity generation with 114.4 GW (26.86%) and 94 GW (22.07%) respectively, in the end, the year 2050 under the reference scenario. This shall be followed by the wind based plants with a total 62.3 GW (14.6%) capacity during the end year. In line with government's vision 2025, the share of nuclear power installed capacity which is 0.78 GW of the total capacity in 2015 is projected to be 12.8 GW by 2050. The installed capacity of solar, hydropower and biomass are only 0.1 GW, 3.961 GW, 0.083 MW respectively in the base year which is optimistically projected to be 54.4 GW, 55.7 GW, 10.3 GW respectively in 2050. It is projected under this scenario that solar, wind and biomass will together generate around 22.1% of overall country's electricity in 2050.

As regards the RET scenario, the documented potential of wind and solar is estimated to be 346 GW and 2,900 GW while another of 60 GW and 3-5GW capacity of hydropower and biomass has been considered respectively. Accordingly, in this scenario, the installed capacity of solar, wind and biomass which are 0.1 GW, 0.3 GW, and 0.1 GW respectively in the base year shall substantially increase to 78 GW (16.8%), 116 GW (25%), and 11 GW (2.37%) respectively by the end year 2050. The capacities of NGCC, hydropower, coal, RoR and nuclear based plant are projected to be 115.9 GW, 59 GW, 40.2 GW, 27.5 GW, and 16.3 GW respectively during the end year. The overall share of renewable energy (inclusive of hydropower) in the installed capacity under this is estimated to be 62.83% in the year 2050.

The total installed capacity in CCM scenario is 416.2 GW which is 9.73 GW less than the REF scenario and 47.7 GW less than the RET scenario in 2050 owing to improved efficiency of Coal based IGCC and Coal based CCS based plants. In this scenario, the coal has a greater share in the overall installed capacity of around 84 GW (20.18%) and is followed by wind energy based plants of 75.5GW (18.14%) and then solar with an installed capacity of 62.3 GW (14.96%) in the year 2050. The capacities of hydropower, NGCC, Coal based IGCC, RoR, Coal based CCS, nuclear and biomass based plants are estimated to be 52.4 GW, 45.9 GW, 30 GW, 26.2 GW, 20 GW, 12.9 GW, and 7 GW respectively in the year 2050.

In the EEC scenario, the total installed capacity is 348.56 GW which is 77.38 GW less than the REF, and also less than other alternative scenarios owing to lower electricity demand and thus reduced generation requirement under this scenario. The installed capacity of coal based plants is projected to be 82 GW (23.52%) followed by NGCC 80.7 GW (23.14%) and then wind based plants of 59.3 GW (17%) in this scenario. The installed capacities of solar, hydropower, RoR, nuclear and biomass based power units in this scenario are estimated to be 50.6 GW, 45.7 GW, 13.8 GW, 8 GW, and 8.4 GW respectively.

It is evident from the results of the generation and installed capacities that divergent set of policy narratives, within different scenarios, would require specific arrangements for the resources and technologies to meet the projected demand.

5.2.3 CO_2 Emissions

The future additions of thermal power plant capacities pose a great challenge to contain CO_2 emission to a minimum level. As per this study, in the year 2015, the CO_2 emissions from power plants of the country were 40.5 million tons, and these are projected to reach 657.17 million tons in 2050 under REF scenario. **Fig. 12** shows CO_2 emissions from the power plants under the REF and alternative scenarios of this study.

In RET scenario, with more than 50% generation from renewable resources, the CO_2 emissions are projected to be minimum and less than the all other scenarios of the study. In this scenario, the CO_2 emissions in the year 2050 are around 400.76 million tons which are 1.64 times less than the REF scenario for the same period.

The CO_2 emissions in CCM scenario are also less than the REF and EEC scenarios due to the utilisation of Coal based IGCC and Coal based CCS technologies. In this scenario, the CO_2 emissions are projected to be 410 million tons of in 2050 which is significantly less than REF and EEC scenarios. Further, in the CCM scenario, this study following Pak-INDC report submitted under the Paris Agreement ensured a reduction of 20% CO_2 emissions by 2030 from the emission level of 2015. This trend has been further continued with 30% reduced CO_2 emission by 2040, and 40% by the year 2050 from the base year CO_2 emissions level. However, this optimistic reduction in CO_2 emission is only subject to required investments and the addition of substantial capacity of renewable energy plants following the year 2020 and onwards. The CO_2 emissions in EEC scenario are 18% less than REF scenario since the reduction in demand requires less electricity generation and thus lower emissions.

5.2.4 Cost Analysis

In this study, the total system cost is expressed in term of NPV which include capital cost, fixed O&M, and variable O&M costs discounted at the rates of 4%, 6%, 8%, 10% during the study period for each of the scenario as shown in **Fig. 13.**

EEC scenario has been estimated with lowest NPV compared to REF, RET and CCM scenarios at all the discount rates. The NPV at the discount rate of 6% which is close to policy rate of State Bank of Pakistan (5.75%), the EEC scenario has the lowest value due to a significant saving on account of lower installed capacity requirement. The total savings estimated under EEC scenario are, as such, estimated to be USD 49.33 billion when compared to NPV of REF scenario, $105 billion saving when compared to NPV of RET scenario and a saving of USD 74.43 billion when compared to NPV of CCM scenario at the discount rate of 6%.

The comparison of NPV of all the scenarios suggests that EEC scenario is the least-cost electricity generation pathway at all the discount rates. The CCM scenario too has lower NPV compared to the RET scenario since capacity requirements of CCM scenario are less than the RET scenario. However, CCM scenario has higher NPV at all the discount rates when compared with the REF scenario since the advanced technologies of CCM scenario shall be capital intensive.

The NPV at all discount rates for REF scenario is less than all other scenarios except that of EEC scenario since mostly conventional technologies are used for power generation in the REF scenario with no focus on reducing the CO_2 emission.

6. Discussion

This exploratory study attempts to provide a judicious description of a possible future state of the electricity demand-supply situation of Pakistan. Supply side scenarios are not merely a prediction; rather, each of four scenarios is one alternative image of what the future can hold for Pakistan's power sector in terms of generation, installed capacity, CO_2 emissions, and cost. The summary results of the study demonstrate that:

- The forecasted electricity demand for the study period is modest and comparable with other studies
- REF scenario fails completely to meet electricity demand by sustainable means of electricity generation, and in providing a cost-effective response to meet INDCs.
- RET and CCM scenario although cost intensive but efficiently reduces CO_2 emissions by 1.64 and 1.6 times respectively less than the REF scenario for the same period.
- However, RET scenario would require more installed capacity due to lower capacity factor and availability issues whereas, CCM requires less installed capacity due to improved efficiencies of modern technologies.
- EEC scenario seems to be the most suitable pathway subject to energy efficiency and conservation measures undertaken. As such, this on scenario would require lower generation and installed capacity, therefore, lowered costs due to reduced demand.
- However, CO_2 emissions under EEC scenario are higher than the RET and CCM scenario.

It is apparent that the EEC scenario seems to perform well due to anticipated utilization of energy efficient appliances, modern technologies and energy conservation measures thus electricity demand is reduced by 20% than other three scenarios. Since REF, RET and CCM scenarios are driven by similar demand-side assumptions, therefore, demand forecast for these scenarios is same; 1706.3 TWh for the end year 2050. The most feasible EEC, though very suitable, yet require serious efforts from the Government to ensure the promotion of energy efficient devices and capacity building towards the energy conservation. EEC scenario implementation would, therefore, also require certain investments towards energy efficiency and conservation. In this context, the domestic sector, being leading electricity consumer group in the base and end year followed by the industrial sector are two important candidates to undertake therein energy efficiency and conservation measures. Despite such initiative and given the past performance on EEC measures in the country, it is challenging to achieve and maintain energy efficiency and conservation although it can even address electricity shortfall and climate change issues in short-term as evident from experience of many developing countries. However, as revealed from the results of this study, the implementation of EEC scenario would be financially suitable as it is estimated that every unit of energy saved would reduce the need for the capacity addition by 2 to 2.5 times. On the other hand, one key shortcoming of EEC scenario is higher CO_2 emissions of 539.2 million tons which are just next to REF scenario and make it an unfitting option in term of meeting the national climate change targets and adhering the international treaties. In terms of CO_2 emissions, the RET scenario has the lowest emission of 400.76 million tons with around 53% generation from indigenous renewable energy resources. As such, this scenario may not only help to reduce CO_2 emissions but also help meeting the submitted INDC's under Paris Agreement as well as achieving Sustainable Development Goals (SDGs), However, the NPV of RET scenario at a discount rate of 6% is highest of other scenarios of the study. This is mainly owing to higher costs of renewable energy conversion technologies, which are now in declining trend, and this scenario too requiring more installed capacity due to lower efficiency of these plants and seasonal variations.

Given the above situation, a combination of EEC and RET scenario suits most the economic and environmental requirements of Pakistan. This arrangement can provide Pakistan with an opportunity to benefit from Carbon offsetting thus earning the carbon credit which is increasingly a popular way of tackling the climate change. Carbon offset purchaser aims to compensate for – or "offset" – their emissions by paying someone else to reduce GHG emissions elsewhere. If GoP fails to penetrate more renewable energy potential in the national grid, another alternate pathway given is CCM scenario, which is also cost-intensive due to use of advanced technologies and conventional fuels. Coal based IGCC and CCS technologies proposed in this scenario shall greatly help in reducing CO_2 emissions thus making Pakistan capable of meeting INDCs s and may also benefit from Carbon offsetting. As regards the REF scenario, following this path Pakistan will further move towards un-sustainability and slower economic growth due to unstable fossil fuel availability and varying prices in the international market and finally it will never be possible to meet any INDCs or SDGs under this scenario with highest projected CO_2 emissions to reach 657.17 million tons in 2050. As per NPV analysis, EEC scenario will save 49.33 billion US dollars when compared to REF scenario, 105 billion

US dollars when compared to RET scenario and 74.43 billion US dollars when compared to CCM scenario at a discount rate of 6%.

In the light of the above discussion of results, it emerges that in order to maintain sustainable electricity supplies to meet the demand with optimal investment and reduced emissions, Pakistan should undertake energy efficiency and conservation measures as well as increase renewable energy share in the overall energy mix of the power generation in the long run.

7. Conclusions

This exploratory study focused on forecasting and balancing long-term electricity demand-supply situation by taking into account the techno-economic and environmental parameters. The LEAP model results of this study are comprehensible, internally consistent and reasonable description of a possible future state of the electricity demand-supply situation of Pakistan. Supply-side scenarios are not merely a prediction; rather, each of four scenarios is one alternative image of what the future can hold for Pakistan's power sector in terms of generation, installed capacity, CO_2 emissions, and cost perspective. The forecasted electricity demand for the study period is modest and comparable with other studies. The overall projected electricity demand of Pakistan in 2050 has been forecasted to be 1706.3 TWh which was only 90.4 TWh in 2015. As such, a growth 19 times higher than the base year demand with an estimated annual average growth of 8.35% over the study period. In case of EEC scenario, the projected electricity demand is estimated to be 1373.2 TWh, which is 20% less than the demand forecasted under REF scenario.

All the supply side scenarios were developed to meet the electricity demand forecast for both the REF (1706.3 TWh) and EEC scenarios (1373.2 TWh) in 2050. As such, EEC scenario required less installed capacity requirement of 354.35 GW followed by CCM, REF, and RET scenarios with installed capacities of 416.21 GW, 425.94 GW, 463.9 GW respectively in 2050. In terms of emission, the REF scenario GHGs from power generation is highest followed by EEC, CCM, and RET scenario. However, cost analysis in terms of NPV suggests that EEC scenario would require the least investment cost followed by REF, CCM, RET scenario.

The study results and analysis are significant and imply that combined EEC and RET scenario would suit most the economic and environmental requirements of Pakistan. As such meeting the reduced electricity demand of EEC scenario by an energy mix as suggested in RET scenario. Following this pathway, therefore, would instantly help in minimizing the with the 5 GW electricity shortfalls at hand. Further, taking advantage of them with the recent creation of MOE, Pakistan should immediately consider developing short term and long term IEP and implement the same forthwith. As per the results of this study, it is evident that following the government's current plans and policies, Pakistan will never become a sustainable economy as it will continue to rely on imported fuels.

However, by following suggested measures, Pakistan's power sector fuel mix can be turned independent of imports thus reducing both the cost and emissions.

In this context, application of energy modeling both at government and academia level for energy system analyses and policy formulation should be undertaken to test various hypothesis and policy perspective. However, modeling results would not provide itself with the policies instead guide decision makers to devise the appropriate path in the formulation of energy policies. Likewise, based on this study, the following policy recommendations are made:

- The institutional structure of the power sector should be optimally rationalized and finalized with due capacity building efforts.
- Energy modeling tools should be substantially used for power policy and expansion planning.
- A strong collaboration between academia and policy making government departments be established so that they may work in close liaison.
- RET technologies development and indigenization through technology transfer should be initiated by AEDB.
- Provinces should be tasked to develop and achieve the energy efficiency and conservation targets.
- Implementation of labeling energy-saving codes for electric appliances should be ensured
- NEECA should be authorized to provide financial incentives to energy conserving consumers.
- Promote energy audits through energy service companies.
- Raise awareness among the public to save energy on a regular basis.

It is anticipated that considering the above recommendation and undertaking the IEP development endeavour towards power policy formulation would yield programs and projects which can bring energy security for Pakistan.

8. Acknowledgment

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References

1. Kreft, S., D. Eckstein, and I. Melchior, *Global Climate Risk Index 2017: Who Suffers Most From Extreme Weather Events? Weather-related Loss Events in 2015 and 1996 to 2015*. 2016: Germanwatch Nord-Süd Initiative eV.
2. Mirjat, N.H., et al., *A review of energy and power planning and policies of Pakistan.* Renewable and Sustainable Energy Reviews, 2017. **79**: p. 110-127.
3. Valasai, G., et al., *Assessment of Renewable Energy for Electricity Generation: Using Pakistan TIMES Energy Model.* Sindh University Research Journal-SURJ (Science Series), 2016. **48**(4): p. 775-778.

4. Kugelman, M., *Pakistan's Interminable Energy Crisis : Is there any way out?* 2015: Asia Program, Wilson Center International Center for Scholars, Washington DC, USA.
5. Bouille, D., *Developing Countries & Global Climate Change: Electric Power Options in Argentina*. 2000: Pew Center on Global Climate Change.
6. Conzelmann, G. and V. Koritarov, *Turkey energy and environmental review. Argonne National Laboratory; 2002*. 2002
7. Mirsagedis, S., et al. *Longterm GHG emissions outlook for Greece*. in *Proceedings of the 6th IAEE European conference on modelling in energy economics and policy. Zurich, Switzerland*. 2004. Citeseer.
8. Kumar, J.A. and C. Radhakrishna. *Sustainable Energy Future by AD2030-India Case Study*. in *Energy 2030 Conference, 2008. ENERGY 2008. IEEE*. 2008. IEEE.
9. Giatrakos, G.P., T.D. Tsoutsos, and N. Zografakis, *Sustainable power planning for the island of Crete.* Energy policy, 2009. **37**(4): p. 1222-1238.
10. Ferreira, P.V. and M.M.T.d. Araújo, *An integrated framework to support sustainable electricity planning.* 2012.
11. Roinioti, A., C. Koroneos, and I. Wangensteen, *Modeling the Greek energy system: Scenarios of clean energy use and their implications.* Energy Policy, 2012. **50**: p. 711-722.
12. Amirnekooei, K., M. Ardehali, and A. Sadri, *Integrated resource planning for Iran: Development of reference energy system, forecast, and long-term energy-environment plan.* Energy, 2012. **46**(1): p. 374-385.
13. McPherson, M. and B. Karney, *Long-term scenario alternatives and their implications: LEAP model application of Panama's electricity sector.* Energy Policy, 2014. **68**: p. 146-157.
14. Von Hippel D, E.P., Tempest K, and L. M., *Strategies for Development of Green Energy Systems in Mongolia Final Report*. 2014.
15. Herdinie, S.S. and E. Sartono, *THE ROLE OF NUCLEAR POWER AND OTHER ENERGY OPTIONS IN COMPETITIVE ELECTRICITY MARKET STUDY USING MESSAGE MODEL.* Jurnal Pengembangan Energi Nuklir, 2003. **5**(1).
16. Saradhi, I., G. Pandit, and V. Puranik, *Energy Supply, Demand and Environmental Analysis–A Case Study of Indian Energy Scenario.* International Journal of Environmental Science and Engineering, 2009. **3**: p. 115-120.
17. Hainoun, A., M.S. Aldin, and S. Almoustafa, *Formulating an optimal long-term energy supply strategy for Syria using MESSAGE model.* Energy policy, 2010. **38**(4): p. 1701-1714.
18. Kumar, M., et al., *Energy Sector Development for 2010-2050 using Message Model.* 2011.
19. Fairuz, S., et al., *Long term strategy for electricity generation in Peninsular Malaysia–Analysis of cost and carbon footprint using MESSAGE.* Energy Policy, 2013. **62**: p. 493-502.
20. Kichonge, B., G.R. John, and I.S. Mkilaha, *Modelling energy supply options for electricity generations in Tanzania.* Journal of Energy in Southern Africa, 2015. **26**(3): p. 41-57.
21. Chen, W. and Z. Wu, *Study on China's future sustainable energy development strategy using MARKAL model.* JOURNAL-TSINGHUA UNIVERSITY, 2001. **41**(12): p. 103-106.
22. Akinbami, J.-F.K., *Renewable energy resources and technologies in Nigeria: present situation, future prospects and policy framework.* Mitigation and Adaptation Strategies for Global Change, 2001. **6**(2): p. 155-182.
23. Heinrich, G., et al., *Ranking and selection of power expansion alternatives for multiple objectives under uncertainty.* Energy, 2007. **32**(12): p. 2350-2369.
24. Tsai, M.-S. and S.-L. Chang, *Taiwan's 2050 low carbon development roadmap: An evaluation with the MARKAL model.* Renewable and Sustainable Energy Reviews, 2015. **49**: p. 178-191.
25. Mondal, M.A.H., S. Kennedy, and T. Mezher, *Long-term optimization of United Arab Emirates energy future: Policy implications.* Applied Energy, 2014. **114**: p. 466-474.
26. Perwez, U., et al., *The long-term forecast of Pakistan's electricity supply and demand: An application of long range energy alternatives planning.* Energy, 2015. **93**: p. 2423-2435.
27. Rehman, S., et al., *An Integrated Modeling Approach for Forecasting Long-Term Energy Demand in Pakistan.* Energies, 2017. **10**(11).
28. Ishaque, H., *Is it wise to compromise renewable energy future for the sake of expediency? An analysis of Pakistan's long-term electricity generation pathways.* Energy Strategy Reviews, 2017. **17**: p. 6-18.
29. Gul, M. and W.A. Qureshi. *Modeling diversified electricity generation scenarios for Pakistan*. in *Power and Energy Society General Meeting, 2012 IEEE*. 2012. IEEE.
30. Kiani, K., *Circular debt touches Rs800bn: Pepco*, in *Dawn*. 2017.

31. Nayyar Hussain, M.A.U., Khanji Harijan, Gordhan Valasai. *Pakistan's Energy System: Integrated Energy Modelling and Formulation of National Energy Policies*. in *14th International Conference on Sustainable Energy Technologies – SET2015*. 2015. Nottingham, UK.
32. PBS, *6th Population and Housing Census*, G.o.P. Pakistan Bureau of Statistics, Editor. 2017: Islamabad.
33. NTDC, *Power System Statistics*. 2015, National Transmission and Despatch Company.
34. NTDC, *Electricity Demand Forecast Based on Regression Analysis*. 2014, Planning Power, National Transmission and Despatch Company.
35. Wasti, S., *Economic Survey of Pakistan 2014-15.* Islamabad: Government of Pakistan, 2015.
36. NEPRA, *State of Industry Report 2015*. 2015, NEPRA.
37. IEA, *World Energy Outlook 2015*. 2015, International Energy Agency: Paris.
38. Rehman, S., et al., *The Future of Sustainable Energy Production in Pakistan: A System Dynamics-Based Approach for Estimating Hubbert Peaks.* Energies, 2017. **10**(11): p. 23.
39. HDIP, *Pakistan Energy Yearbook 2015*. Hydrocarbon Development Institute of Pakistan (HDIP), 2015.
40. UNDP, *Climate Public Expenditure Institutional Review*. 2015, United Nations Development Programme Pakistan: Islamabad, Pakistan.
41. UNFCCC. *Pakistan's Intended Nationally Determined Contributions (INDC)*. 2016 [cited 2017 October 31]; Available from: www4.unfccc.int/Submissions/INDC/.../Pakistan/1/Pak-INDC.pdf.
42. GOP, *Economic Survey of Pakistan 2015-16*. 2016, Ministry of Finance, Government of Pakistan.
43. EIA, U., *Updated capital cost estimates for utility scale electricity generating plants.* US Energy Inf. Adm, 2013. **524**.
44. WBG, *World Development Indicators | World DataBank*. 2015, World Bank Group.
45. NTDC. *Daily Operational Energy Data*. 2015 [cited 2015 September 15]; Available from: http://www.ntdc.com.pk/doed.php.
46. Government of Pakistan, N., *Status of Industry Report.* 2015.
47. IRG, *Pakistan Integrated Energy Model (Pak-IEM)*. 2010, Asian Development Bank and Ministry of Planning and Development, Government of Pakistan.
48. NTDC, *National Power System Expansion Plan 2011-2030*. 2011, National Transmission and Despatch Company.
49. Farooqui, S.Z., *Prospects of renewables penetration in the energy mix of Pakistan.* Renewable and Sustainable Energy Reviews, 2014. **29**: p. 693-700.
50. ADB, *Sustainable Energy Efficiency Development Program: Consultants' Reports*. 2009, Asian Development Bank.

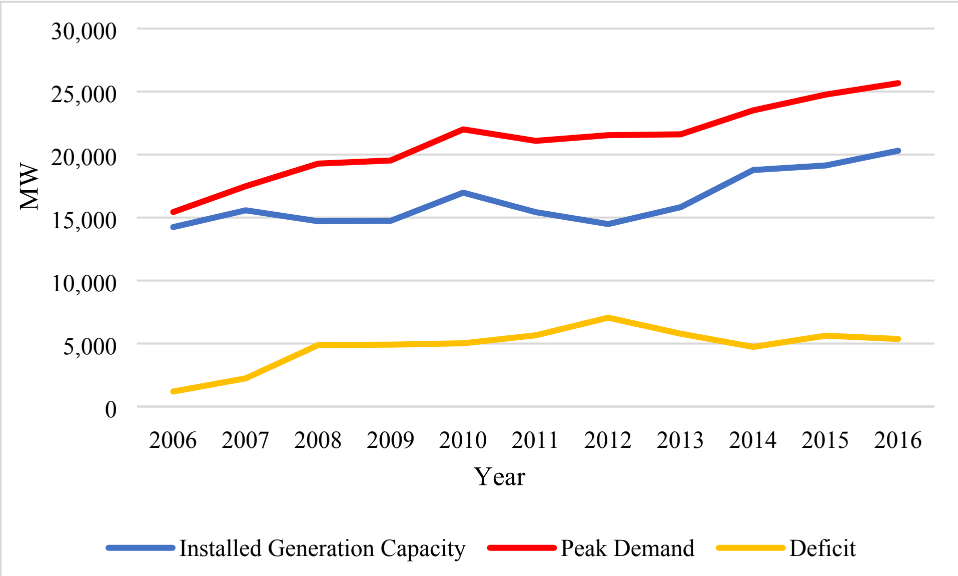

Fig.1. Electricity demand and supply gap [36]

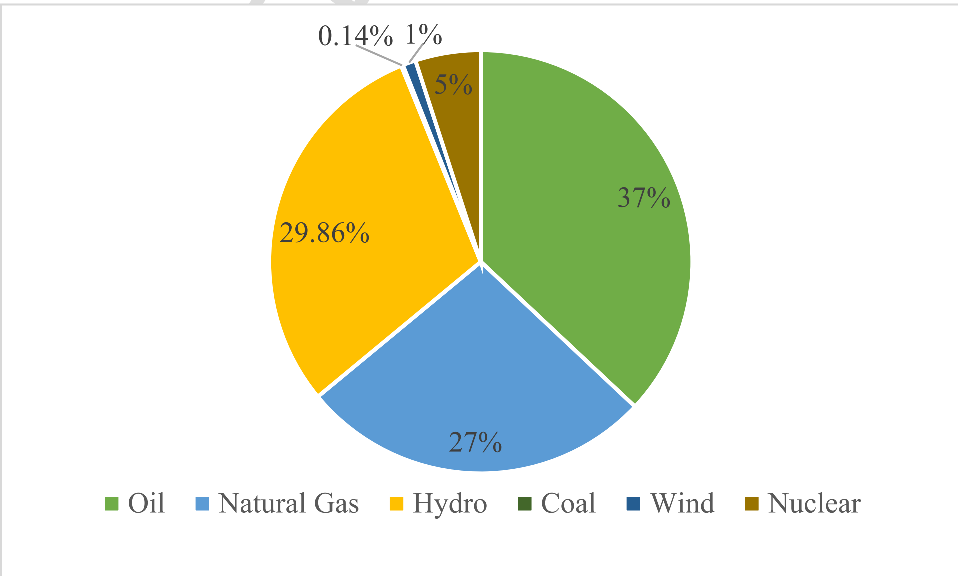

Fig.2. Electricity generated by the source 2015 [39]

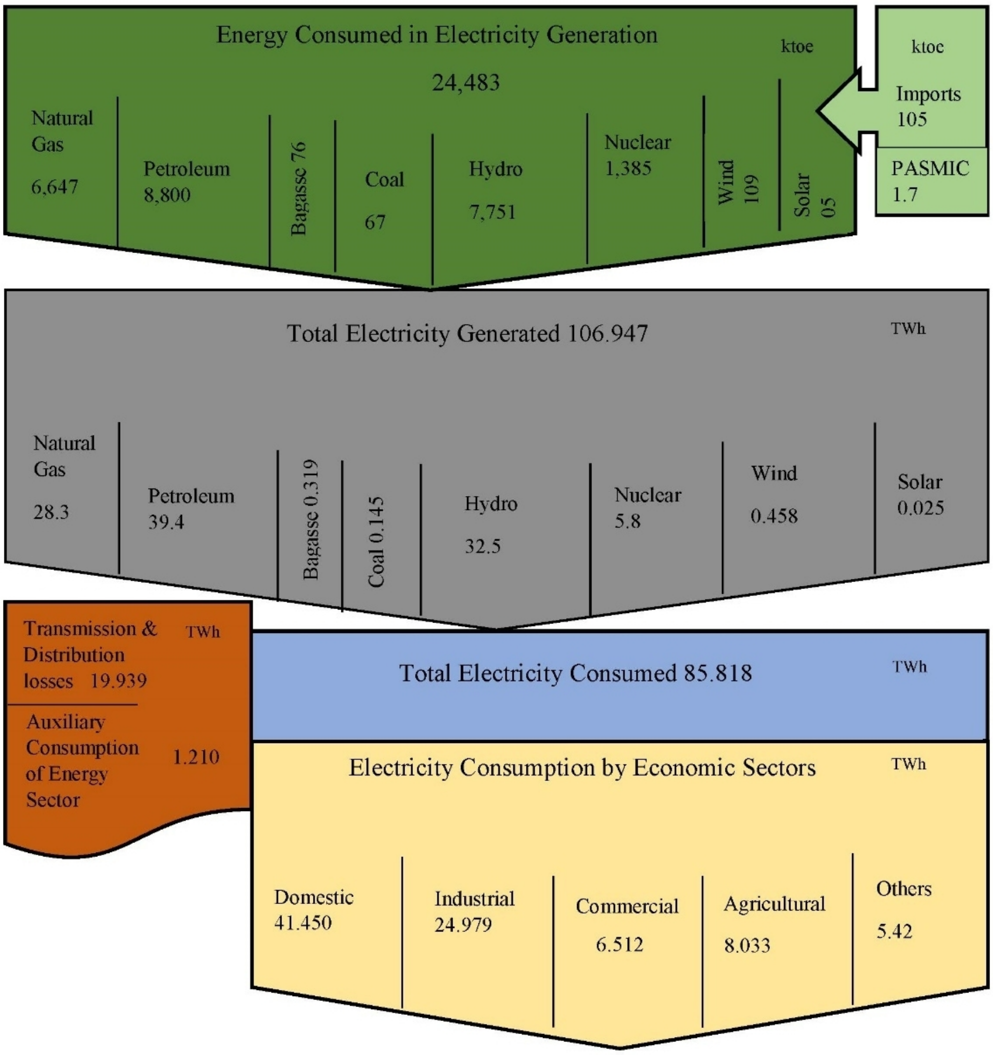

Fig.3. Electricity flowchart of Pakistan-2015

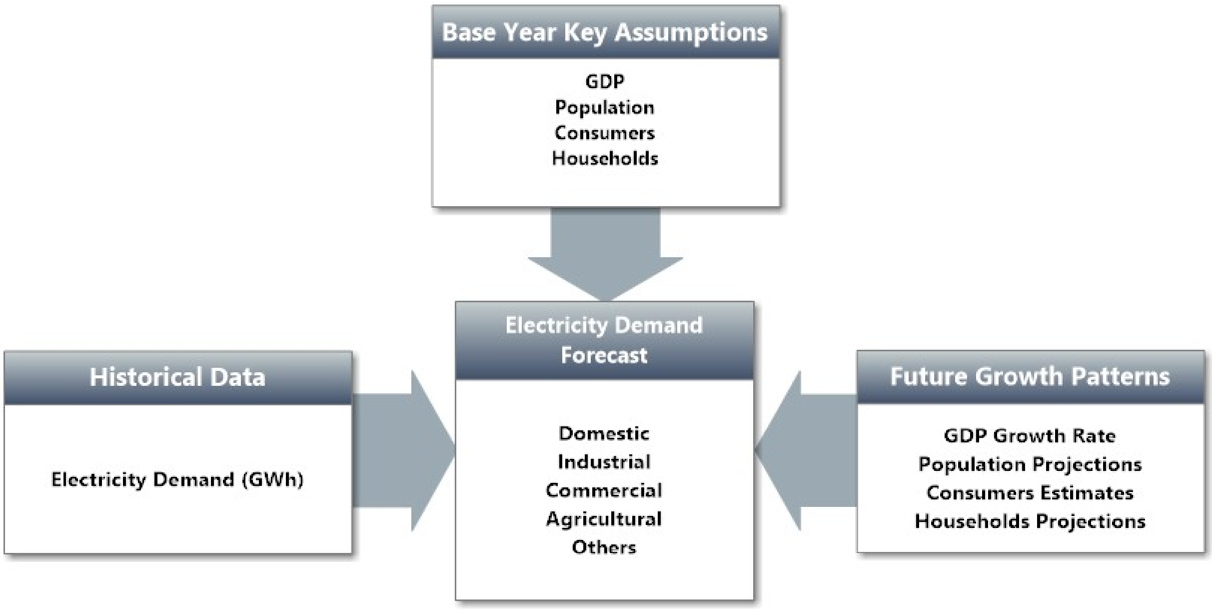

Fig.4. Electricity demand forecast structure of LEAP model

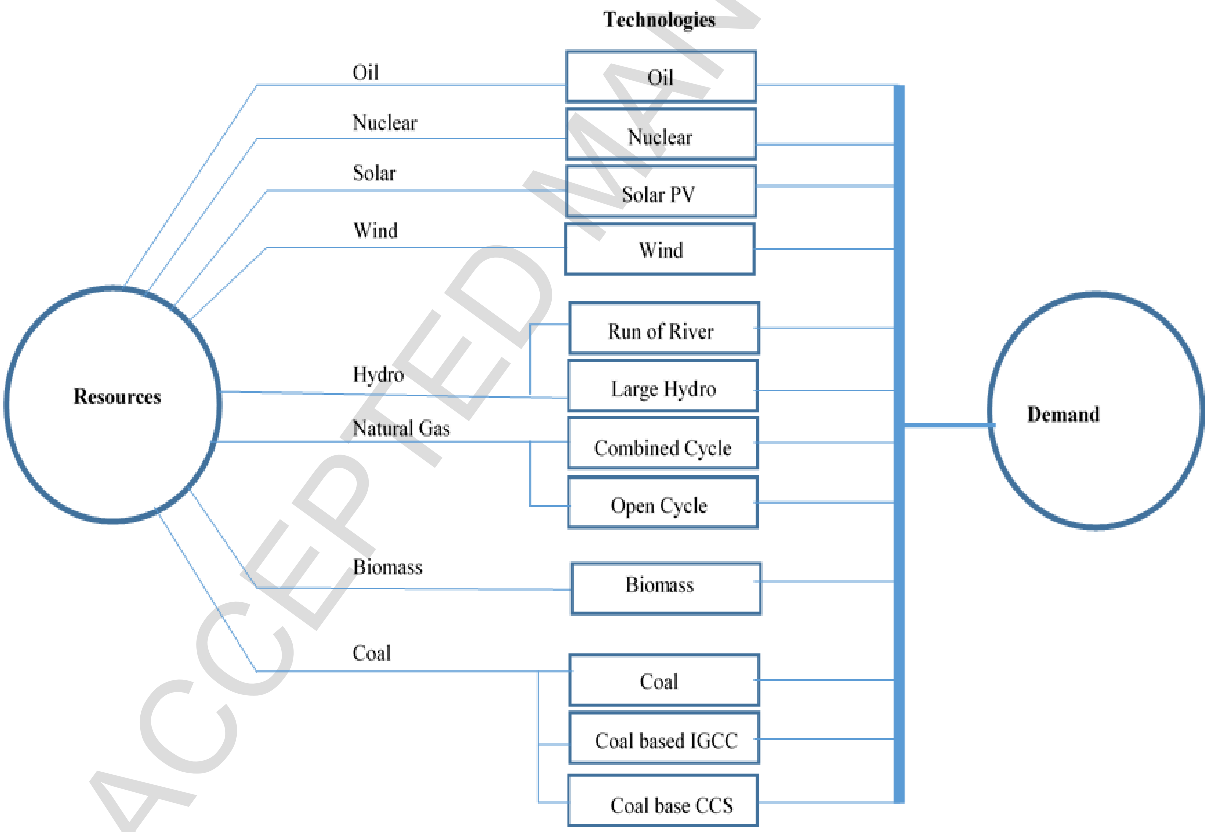

Fig.5. Electricity generation module for Pakistan in LEAP model

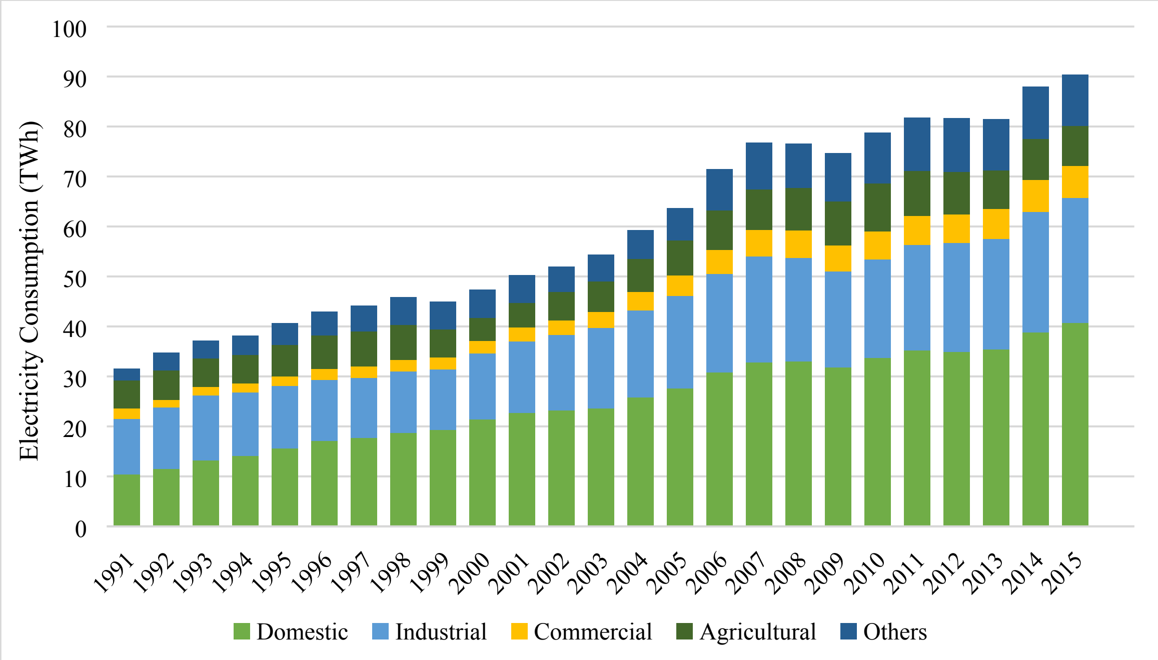

Fig.6. Electricity consumption of Pakistan for the period 1991-2015 [35]

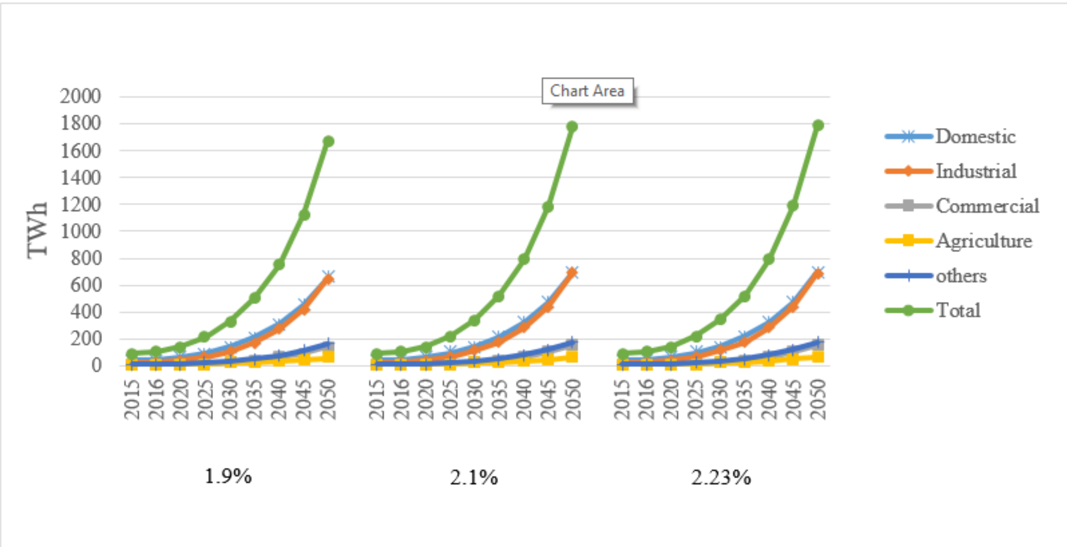

Fig.7. Sensitivity Analysis of electricity demand forecast under different population growth rates

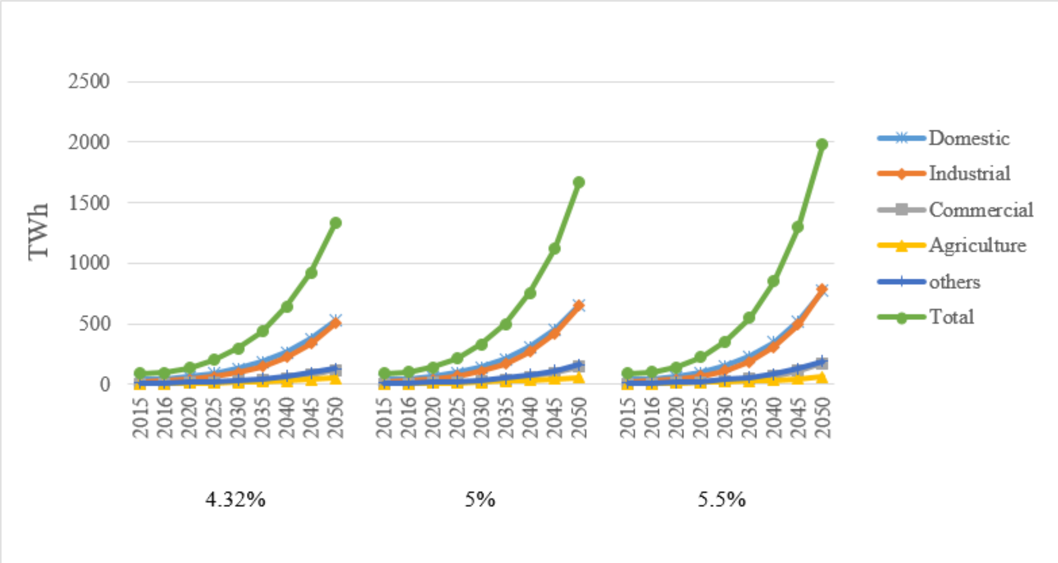

Fig.8. Sensitivity Analysis of electricity demand forecast under different GDP growth rates

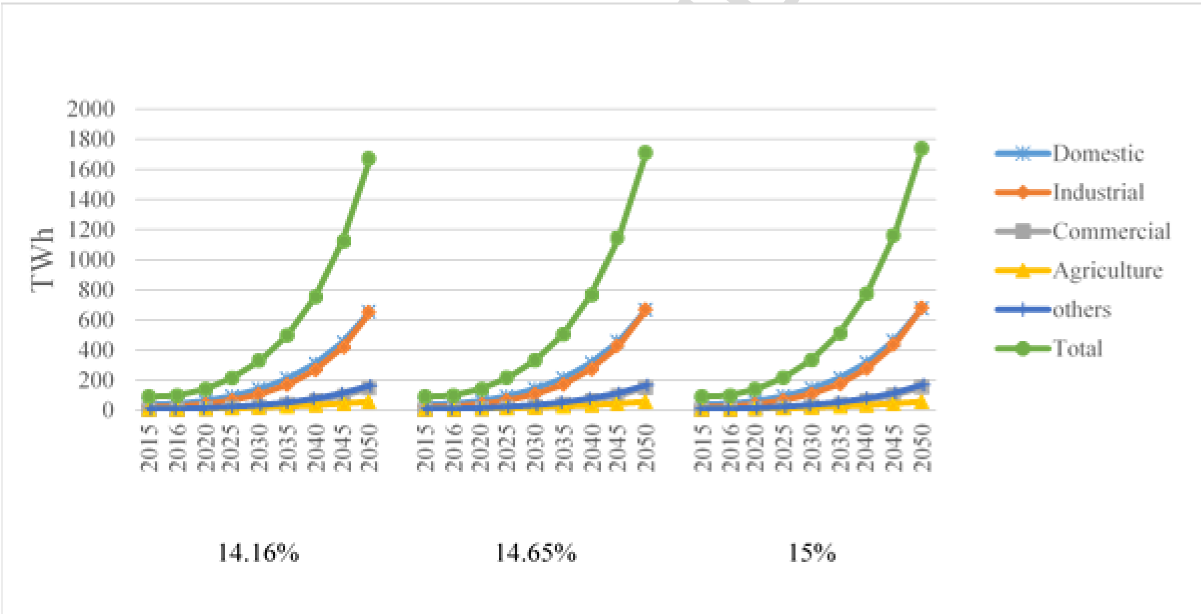

Fig.9. Sensitivity Analysis of electricity demand forecast under different Number of Consumer (NC) growth rates

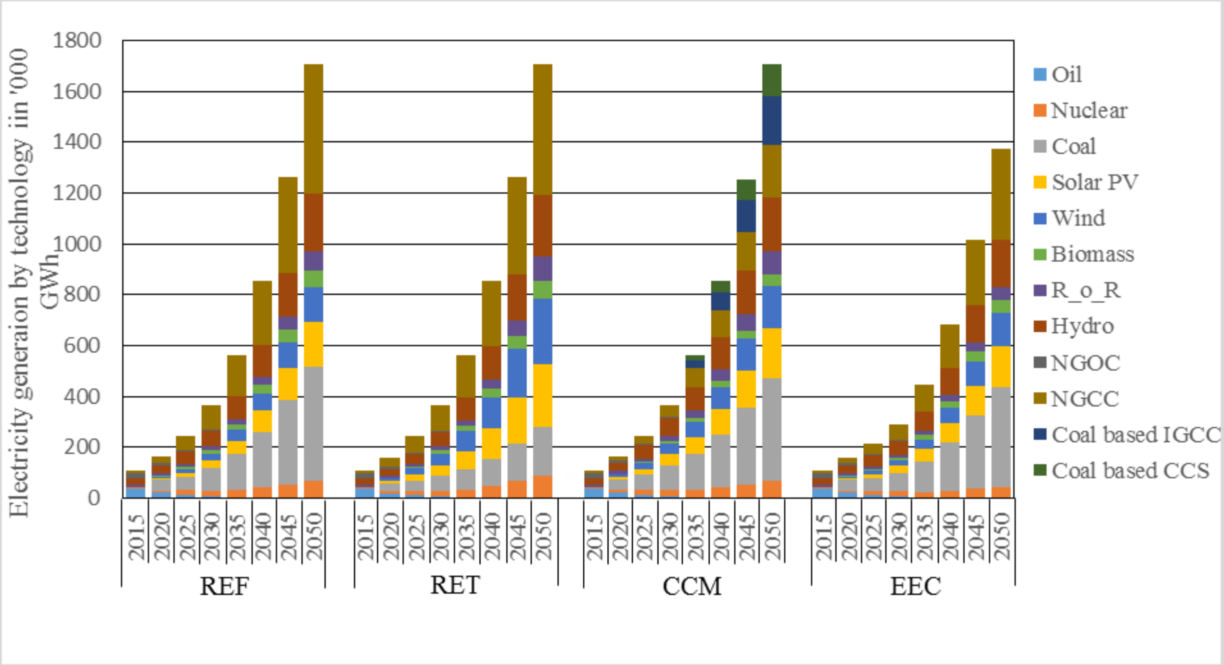

Fig.10. Electricity generation in all scenarios during study period

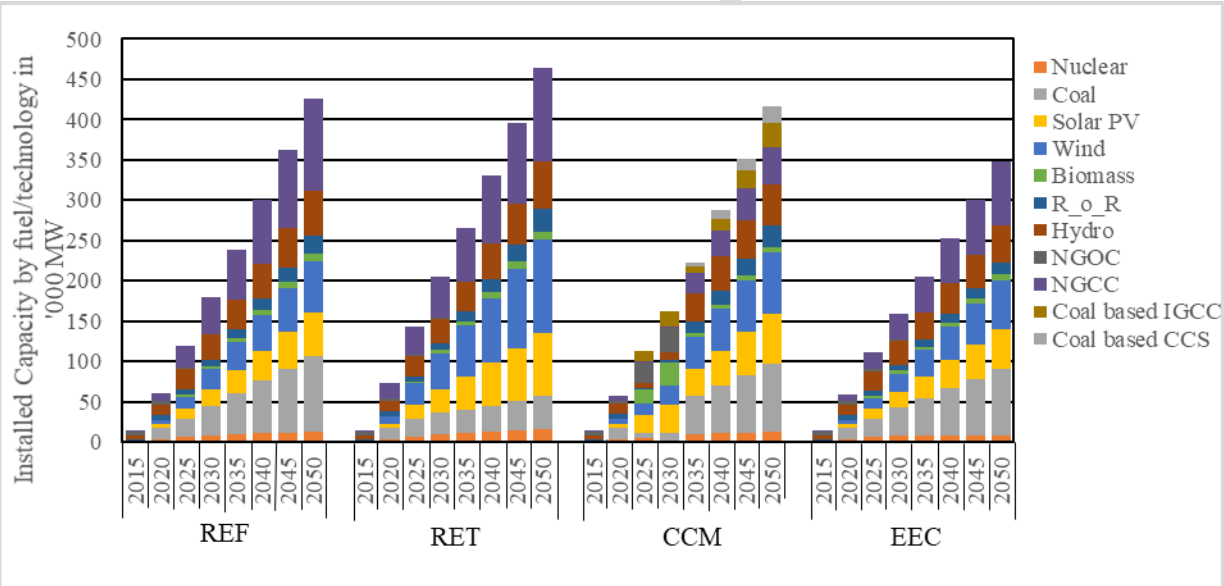

Fig.11. Installed capacity in all scenarios during study period

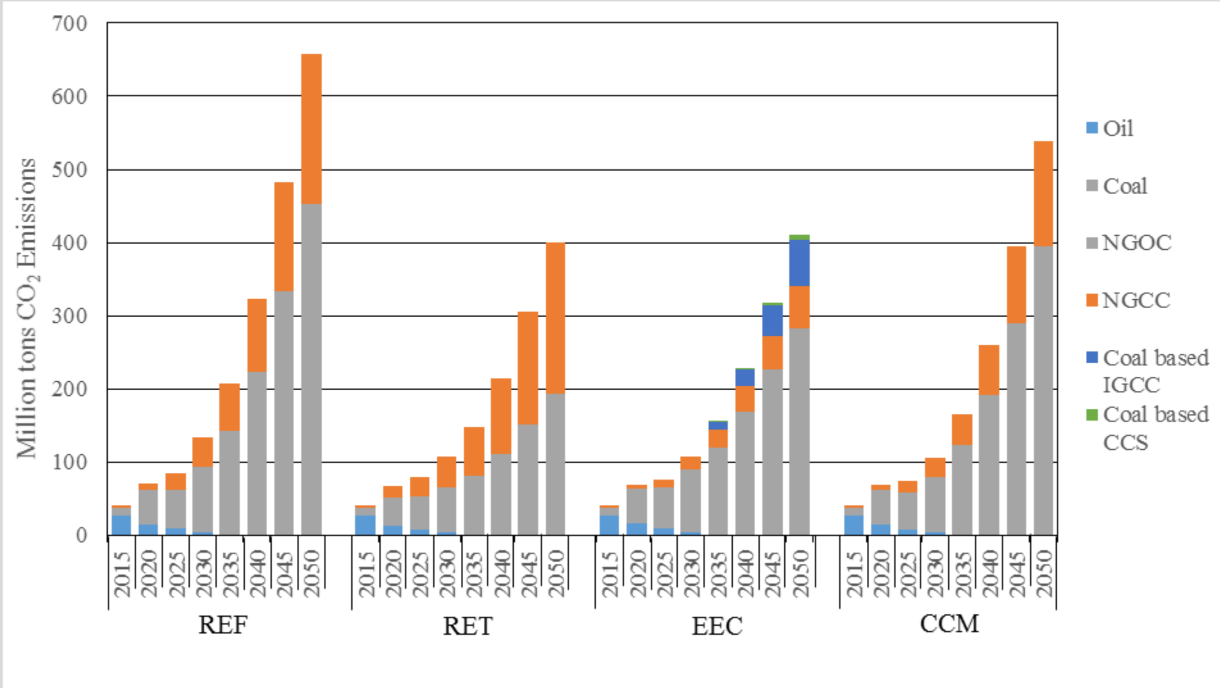

Fig.12. CO_2 emissions in all scenarios during study period

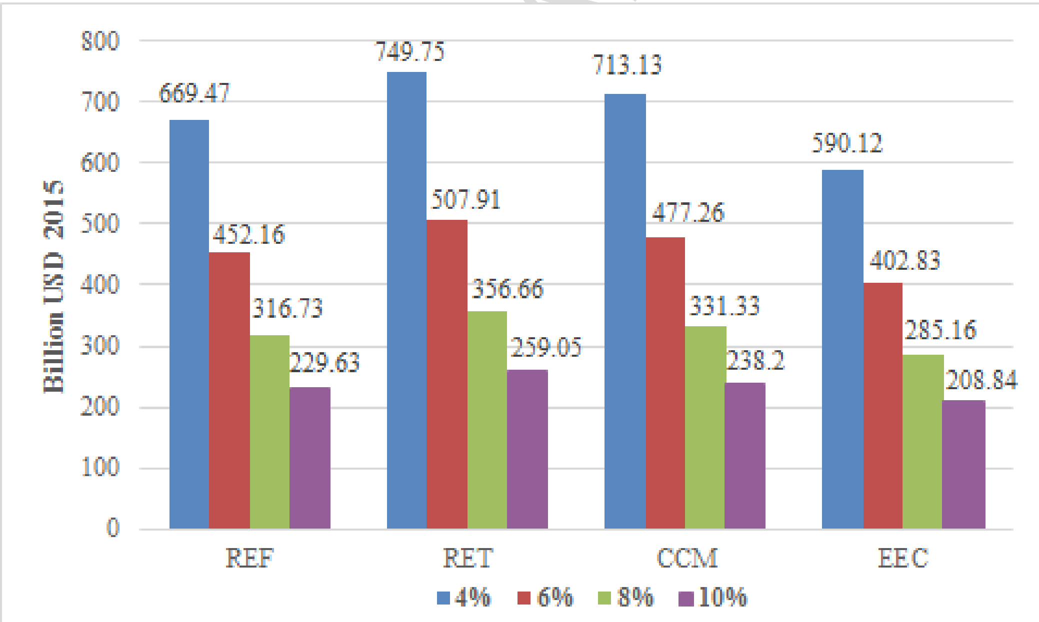

Fig.13. NPV at different discount rates for all scenarios of the study

Highlights

- Pakistan's electricity system modeled for the period of 2015-2050 using LEAP tool
- Electricity demand for Pakistan in 2050 has been forecasted as 1706 TWh
- Energy efficiency & conservation scenario decreases 20% of electricity demand by 2050
- Four supply side scenarios developed to meet both of these forecasted demands
- Role of energy efficiency improvement & renewable energy integration is significant

Table 1:
Various studies undertaken using energy modeling tools [5-25]

Study	Energy Model Tool	Focus of Study	Country
Bouille (2000)	ENPEP BALANCE	Electric Power Options in Argentina	Argentina
Conzelman and Koritarov (2002)		Turkey energy and environmental review	Turkey
Mirsagedis et al. (2004)		Long-Term GHG Emissions Outlook	Greece
Kumar and Radhakrishna (2008)		Sustainable energy future	India
Giatrakos et al. (2009)	LEAP	Sustainable power planning	Crete
Ferreira and Araújo (2012)		An integrated framework to support electricity planning.	Portugal
Roinioti et al. (2012)		energy scenarios for the future – with a focus on the electricity production system	Greek
Amirnekooei et al. (2012)		Integrated Resource Planning for Energy System	Iran
McPherson and Karney (2014)		Long-term scenario alternatives and their implications in Electricity Sector	Panama
Von Hippel D et al. (2014)		Strategies for Development of Green Energy Systems	Magnolia
Herdinie and Sartono (2003)	MESSAGE	Assessment of Role of Nuclear Power and other electricity generation options	Indonesia
Saradhi et al. (2009)		Energy Supply, Demand and Environmental Analysis	India
Hainoun et al. (2010)		Long-term energy supply strategy development	Syria
Kumar et al. (2011)		Energy Sector Development for 2010-2050	Malaysia
Fairuz et al. (2013)		A long-term strategy for electricity generation-Analysis of cost and carbon footprint.	Malaysia
Kichonge et al. (2015)		Modeling energy supply options for electricity generations in Tanzania	Tanzania
Chen and Wu (2001)	MARKAL/ TIMES	China's future sustainable energy development	China
Akinbami (2001)		Renewable energy resources and technologies Assessment and Policy Framework	Nigeria
Heinrich et al. (2007)		Ranking and selection of power expansion alternative	South Africa
Tsai and Chang (2015)		Low carbon energy pathways	Taiwan
Mondal et al. (2014)		Evaluation of future energy-supply strategies for the United Arab Emirates (UAE) power sector	UAE

Table 2:
Summary of past relevant studies and contribution of this study for Pakistan's electricity sector [26-29]

Studies	Studies Period	Demand Projection (TWh)	Supply Side Scenarios	Recommended Scenario	Demand Forecast Consideration			Sensitivity Analysis of Demand Forecast	Supply Side Consideration					
					GDP	Consumers	Population		Energy Policy 2013	CPEC	Vision 2025	50% Renewable Share	CCS & IGCC Technologies	INDCS' under Paris Agreement
Usama et al. (2015)	2011-2030	312 (2030)	– BAU – New Coal – Green Future	GF	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗
Rehman et. al. (2017)	2014-2035	172 (2035)	–	-	✓	✗	✗	✗	–	–	–	–	–	–
Hanan et al. (2017)	2014-2035	303.7 (2035)	– GP – DSM – REN	REN	✓	✗	✓	✗	✓	✓	✓	✗	✗	✗
Maryam et al. (2009)	2008-2030	368 (2030)	– BAU – ORR – HR – HNR	HR and ORR	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗
This Study (2018)	2015-2050	1706.1 (2050)	– REF – RET – CCM – EEC	Combined EEC and REN	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Acronyms:

BAU = Business as Usual

HR= Hydro Resources

NC= New Coal
GF= Green Future
GP= Government Policy
DSM= Demand Side Management
REN= Renewable Energy
ORR= Other Renewable Resources
HNR= Hydro, Nuclear and other Renewable resources
REF= Reference
RET= Renewable Energy Technology
CCM= Clean Coal Maximum
EEC= Energy Efficiency and Conservation

Table 3:
Number of electric consumers in the five demand sectors of Pakistan (million) [35]

Year	Domestic	Industrial	Commercial	Agricultural	Others	Total
2000	10.783	0.224	2.001	0.176	0.009	13.193
2005	13.888	0.233	2.378	0.202	0.010	16.714
2011	18.954	0.293	2.873	0.282	0.013	22.419
2015	21.843	0.335	3.162	0.320	0.014	25.676

Table 4
Installed capacity of supply entity in Pakistan-2015 [39]

Entity	Installed capacity (MW)
GENCOs	4,900
PPIB	8,766
K-Electric	1,875
WAPDA (Hydel)	7,030
PAEC	750
Total	23,321

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Table 5
Various energy/power policies by the Government of Pakistan [2]

Year	Policy	Focus	Institutional Development/Restructuring	Key Features and Targets
1994	Power Policy	Introduced Independent Power Producers (IPPS) and emphasized on energy conservation	Development of Private Power and Infrastructure Board (PPIB)	● Fixed "capacity price" payments to IPPs regardless of electricity generation and immunity from various taxes and surcharges ● ~60% share of hydropower in electricity generation reduced to ~25%
1995	Hydropower Policy	1994 Power Policy rectified for Hydropower Generation (Fiscal incentives and exemption from certain taxes and duties were extended)	-	Despite extended incentives, the policy could not attract private sector investments in the hydropower sector as per policy objectives.
1998	Policy for New Private Independent Power Projects	Amended/Revised 1994 Policy with more rationalization, i.e., introducing competition/bidding.	● The process of restructuring of public giant WAPDA moved further ● Earlier in 1997, independent power sector regulator NEPRA established.	Like its precedents, this policy also continued the exemption from certain taxes and duties such as income tax, customs duties, and sales tax, for importing equipment by IPPs favour thermal power generation. However, contrary to 1994 and 1995, the policy emphasis protecting consumers' and investors' interests and enabling a competitive environment in the market.
2002	Power Generation Policy	This policy encouraged investment from private, public-private and public-sector organizations with features of fuel supply and power purchase agreements	Establishment of AEDB in 2003	The long terms target of power generation of. 20,000 MW by 2015 was set with the introduction of competitive tariffs; an Energy Purchase Price (EPP) and a Capacity Purchase Price (CPP)
2006	Development of Renewable Energy for Power Generation	Focused on the development of Small Hydropower, Wind, and Solar and Biofuel Technologies.	The administrative control of AEDB was transferred to the Ministry of Water and Power	The policy introduced exemptions from duties and taxes for importing required machinery of renewable energy plants. A target of 5% renewable energy share in the overall energy mix of power generation was also set.
2008	National policy for Power Cogeneration by Sugar Industry	This policy envisaged co-generation projects based on bagasse produced in the sugar plants	-	The policy offered a levelized tariff for the co-generation plants with a capacity of at least 60 MW.
2010 - 2012	National Energy Policy	This Policy focused on Energy Conservation and proposed short-term and long-term plans for generation of electricity.	Introduction of Rental Power Plants (RPPs)	This policy emphasized on rehabilitation of existing public-sector power plants
2013	National Power Policy	Focused on the development of electricity generation and energy conservation projects to overcome the energy crisis.	-	The policy envisaged upfront tariff and competitive bidding to lower the cost of the electricity.
2015	Power Generation Policy	Offering enhanced incentives and simplified processing for the investors in order to narrow the demand and supply gap utilizing the indigenous resources mainly with safeguarding the environment and taking all the stakeholders on board.	Upgrading of Climate Change cabinet division to Ministry of Climate change	The policy encouraged the Public-Private Partnership (PPP) by offering incentives

Table 6
Key Assumptions [35, 42]

Key Assumptions	Input Parameters	Growth %	Reference
Population	191.71 Million	1.92	Pakistan economic survey-2015
GDP	257.71 Billion USD	5	NTDC
Industrial GDP	52.31 Billion USD	5.81	NTDC
Agricultural GDP	53.86 Billion USD	2.73	NTDC
Households	30.43 Million		As per expression "Population = Household x Household size," Thus, household increase with an increase in Population

Table 7
Model input parameters

Exogenous input	Endogenous variables
GDP growth trend Electric consumers growth	Electricity demand sector wise (domestic, industry, commercial, agricultural and others)
Fuel cost Lifetime of each technology	Electricity Intensity Generation Capacity (Oil, gas combined cycle And Open cycle, hydro run of river and dam wind, solar)

Table 8
Pakistan's power generation technologies (2015) [37,45-50]

Power Plants	Capacity (MW)	Fuel Consumed (TOE)	Generation (GWh)	Capital Cost (Million $/MW)	Fixed cost (thousand $/MW)	Variable cost ($/MWh)	Efficiency (%)	Maximum Availability (%)
Oil	9212	8,800,432	38690	1.16	26.21	8.523	38.8	88
Nuclear	787	-	4699	5.71	96.27	2.208	34.1	85
Coal	150	67,638	102	3	32.18	4.613	33.0	75
Solar	100	-	25	4	25.48	0	100	18
Wind	256	-	300	2	40.82	0	100	35
Biomass	83	-	319	4	109.01	5.428	35	80
Large hydro	3961	40,606*	22280	1	13.16	4.2	86.5	53
Run of river	2334		10699	7	94.11	7.88	86.5	53
Gas combined cycle	1833.8	6,847,894	9982.72	1	15.86	3.375	54	70
Gas open cycle	4898.2		21213.72	0.698	7.27	10.702	32.7	70
Coal-based IGCC	-	-	-	3.5	87.5	7	45	75
Coal-based CCS	-	-	-	4.2	105	7.7	34	75
Total	23,615	15,756,570	108,310.44	-	-	-	-	-

*(Auxiliary Consumption)

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Table 9
Summary of scenario alternative of this study.

Scenario	Objective	Main Resources
Reference (REF)	In this scenario, the government's current plan and policy is followed.	As per the government plan and policy.
Renewable Energy Technologies (RET)	Under this scenario, the renewable energy resources and technologies are preferred.	Renewable energy resources, hydro, solar, wind, and biomass
Clean Coal Maximum (CCM)	Under this scenario share of clean coal is maximum and is preferred.	Indigenous coal, renewables energy, natural gas and nuclear.
Energy Efficiency and Conservation (EEC)	Under this scenario, the efficiency improvement and conservation measures are considered.	Efficiency measures and conservation potential

Table 10
Sector-wise forecasted electricity demand (TWh)

Economic Sector	2015	2030	2035	2050
Domestic	40.8	141.8	212.2	667
Industrial	24.9	106.7	171.1	662.1
Commercial	6.4	26.1	41.0	149.3
Agricultural	8	19.9	26.8	60
Others	10.2	35.6	53.3	167.7
Total	**90.3**	**330.1**	**504.4**	**1706.3**

Table 11
Sector-wise forecasted number of electric consumers (Millions)

Consumer Groups	2015	2030	2035	2050
Domestic	21.84	29.03	31.58	39.71
Industrial	0.34	0.49	0.56	0.78
Commercial	3.16	4.93	5.66	8.32
Agricultural	0.32	0.59	0.72	1.27
Others	0.01	0.02	0.02	0.03
Total	**25.67**	**35.06**	**38.54**	**50.11**